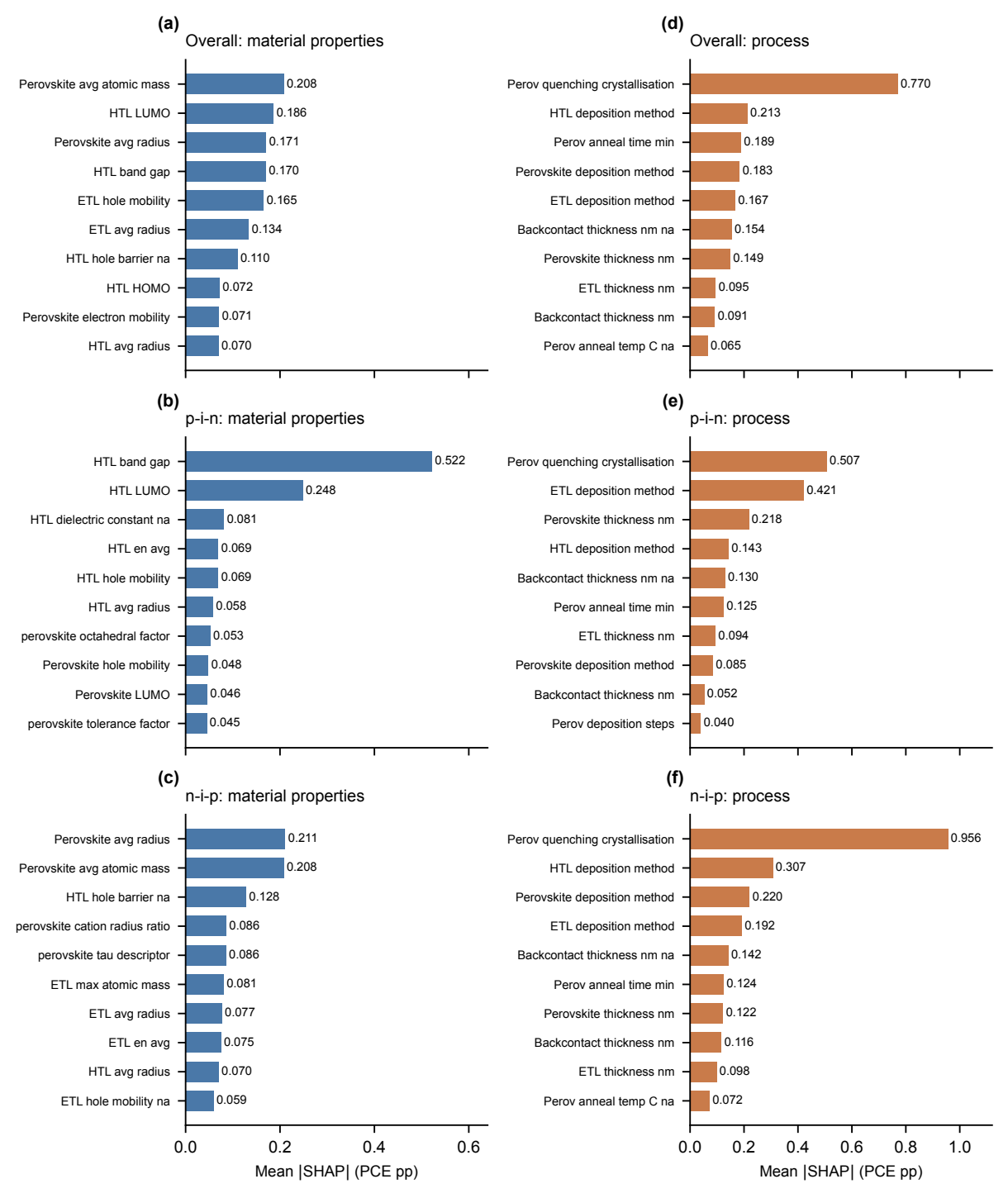
(a)
Overall: material properties
Perovskite avg atomic mass 0.208
HTL LUMO 0.186
Perovskite avg radius 0.171
HTL band gap 0.170
ETL hole mobility 0.165
ETL avg radius 0.134
HTL hole barrier na 0.110
HTL HOMO 0.072
Perovskite electron mobility 0.071
HTL avg radius 0.070
(b)
p-i-n: material properties
HTL band gap 0.522
HTL LUMO 0.248
HTL dielectric constant na 0.081
HTL en avg 0.069
HTL hole mobility 0.069
HTL avg radius 0.058
perovskite octahedral factor 0.053
Perovskite hole mobility 0.048
Perovskite LUMO 0.046
perovskite tolerance factor 0.045
(c)
n-i-p: material properties
Perovskite avg radius 0.211
Perovskite avg atomic mass 0.208
HTL hole barrier na 0.128
perovskite cation radius ratio 0.086
perovskite tau descriptor 0.086
ETL max atomic mass 0.081
ETL avg radius 0.077
ETL en avg 0.075
HTL avg radius 0.070
ETL hole mobility na 0.059
0.0 0.2 0.4 0.6
Mean |SHAP| (PCE pp)
(d)
Overall: process
Perov quenching crystallisation 0.770
HTL deposition method 0.213
Perov anneal time min 0.189
Perovskite deposition method 0.183
ETL deposition method 0.167
Backcontact thickness nm na 0.154
Perovskite thickness nm 0.149
ETL thickness nm 0.095
Backcontact thickness nm 0.091
Perov anneal temp C na 0.065
(e)
p-i-n: process
Perov quenching crystallisation 0.507
ETL deposition method 0.421
Perovskite thickness nm 0.218
HTL deposition method 0.143
Backcontact thickness nm na 0.130
Perov anneal time min 0.125
ETL thickness nm 0.094
Perovskite deposition method 0.085
Backcontact thickness nm 0.052
Perov deposition steps 0.040
(f)
n-i-p: process
Perov quenching crystallisation 0.956
HTL deposition method 0.307
Perovskite deposition method 0.220
ETL deposition method 0.192
Backcontact thickness nm na 0.142
Perov anneal time min 0.124
Perovskite thickness nm 0.122
Backcontact thickness nm 0.116
ETL thickness nm 0.098
Perov anneal temp C na 0.072
0.0 0.2 0.4 0.6 0.8 1.0
Mean |SHAP| (PCE pp)